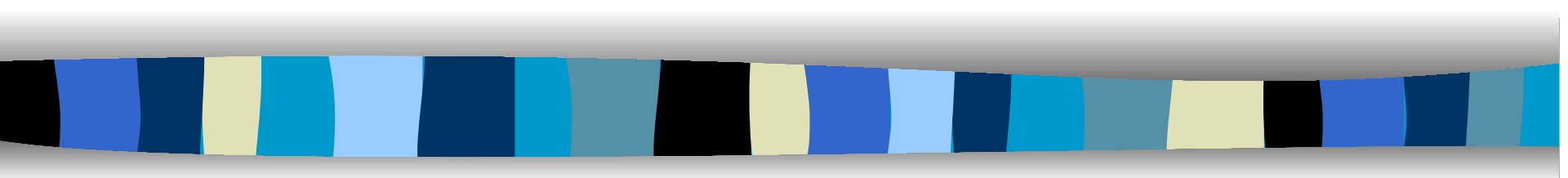


Symmetric Cryptography



Modes of Operation

Modes of operation

- Block cipher uses same key to different blocks of plaintext.
- This is not good for security.
- So, to generate strong and secure ciphertext NIST defines five modes of operations of block ciphers.
- A mode of operation is a technique for enhancing the effect of a cryptographic algorithm.

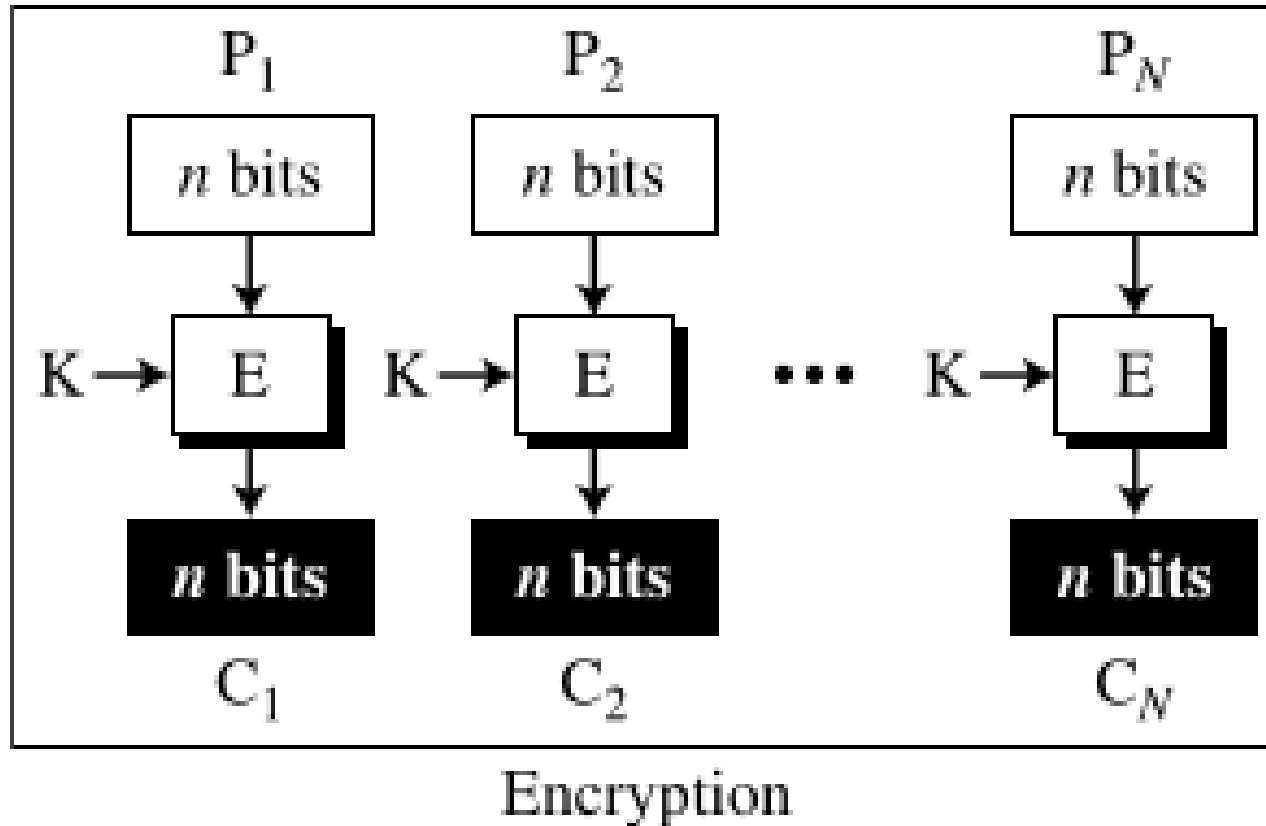
Modes of operation

- Five modes of operations are
- Electronic Codebook (ECB)
- Cipher Block Chaining (CBC)
- Cipher Feedback (CFB)
- Output Feedback (OFB)
- Counter Mode (CTR)

Electronic Codebook

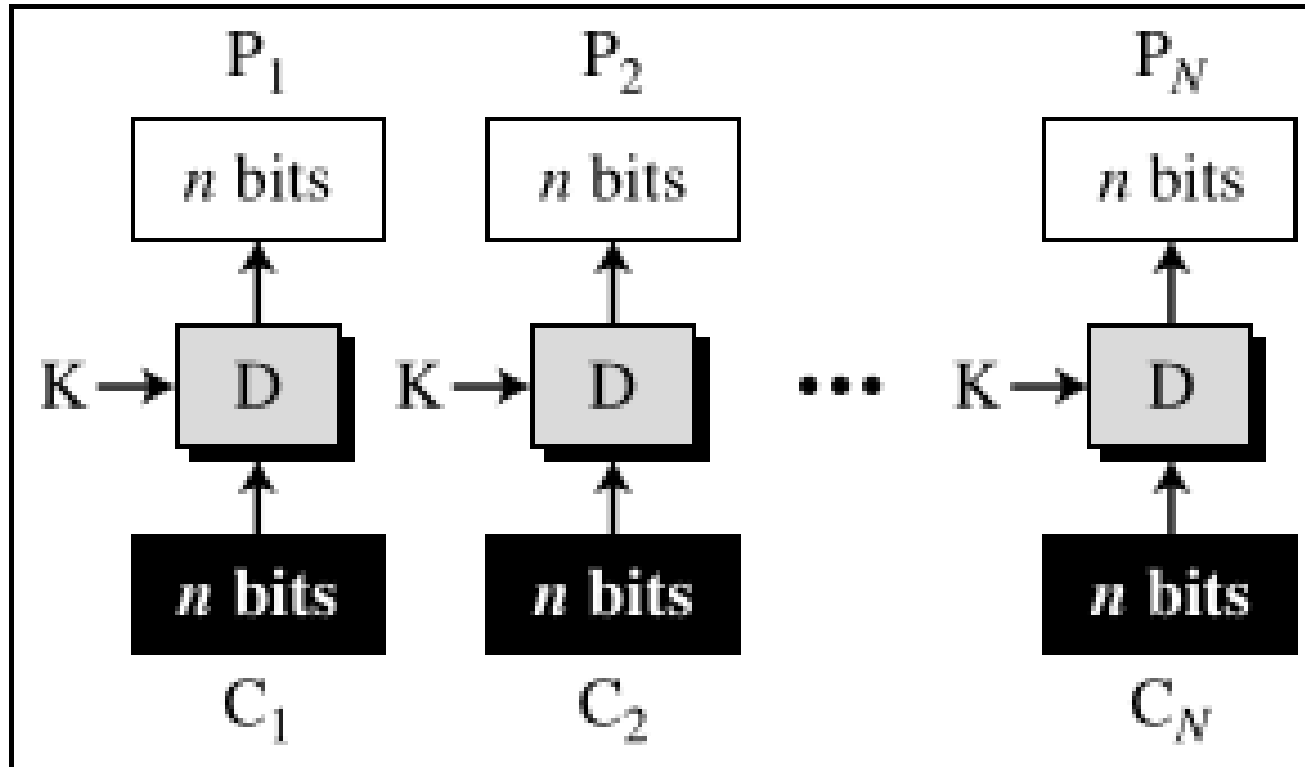
- The simplest mode of operation is called the electronic codebook (ECB) mode.
- The plaintext is divided into different blocks of b -bits.
- If the last block is not of b -bits, then block is padded
- The same key is used to encrypt and decrypt each block.

Electronic Codebook



- E is block encryption algorithm.
- E can be either DES or AES

Electronic Codebook



Decryption

- D is block decryption algorithm.
- D can be either DES or AES

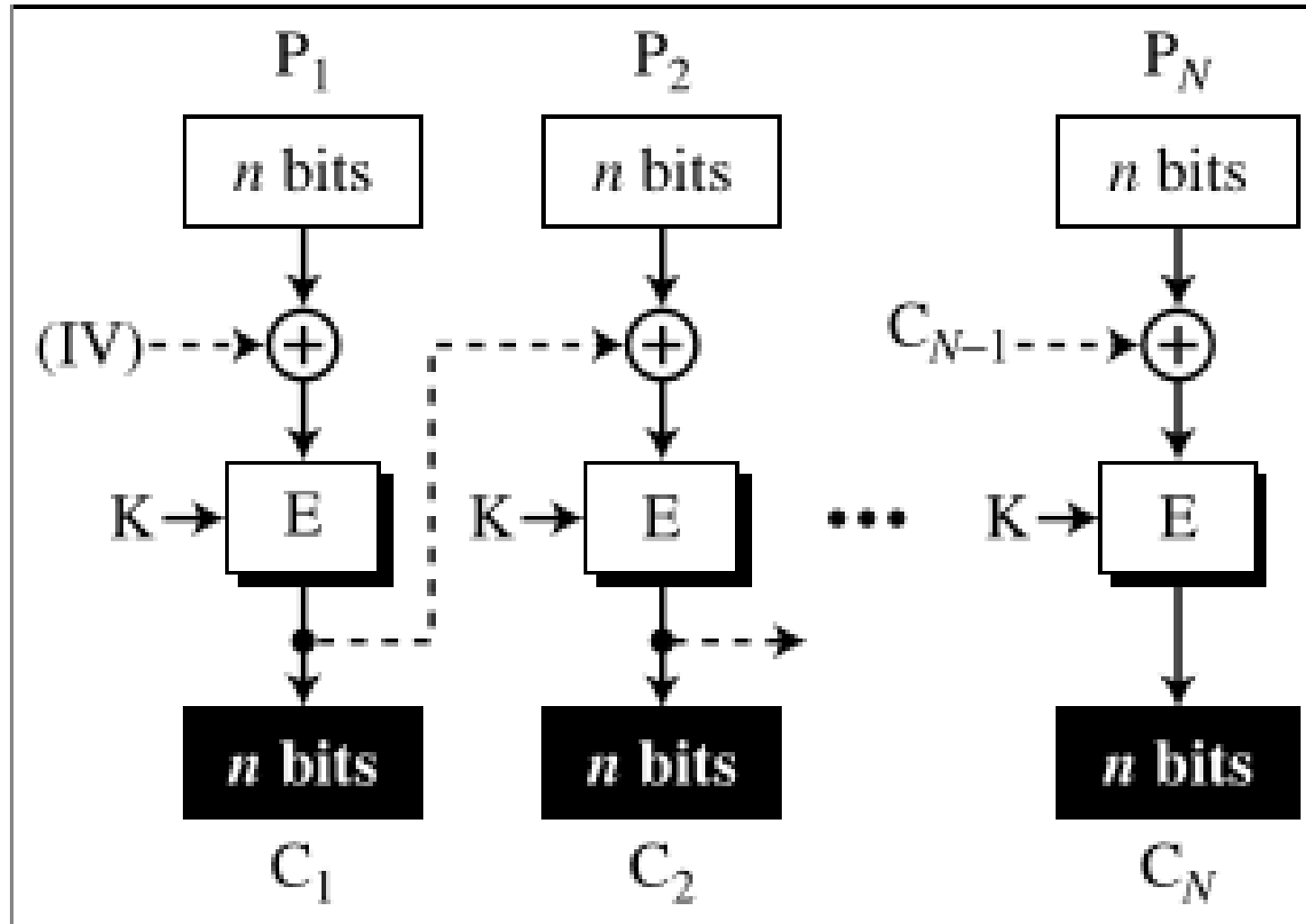
ECB limitations

- If a message has two identical blocks, the ciphertext will be two identical blocks
- Blocks can be rearranged by an adversary to his advantage
- Message information is not sufficiently diffused
- A single bit error can create errors in several bits in corresponding block.
- Other blocks are not affected.
- Using the same key on multiple blocks makes it easier to break

Cipher Block Chaining

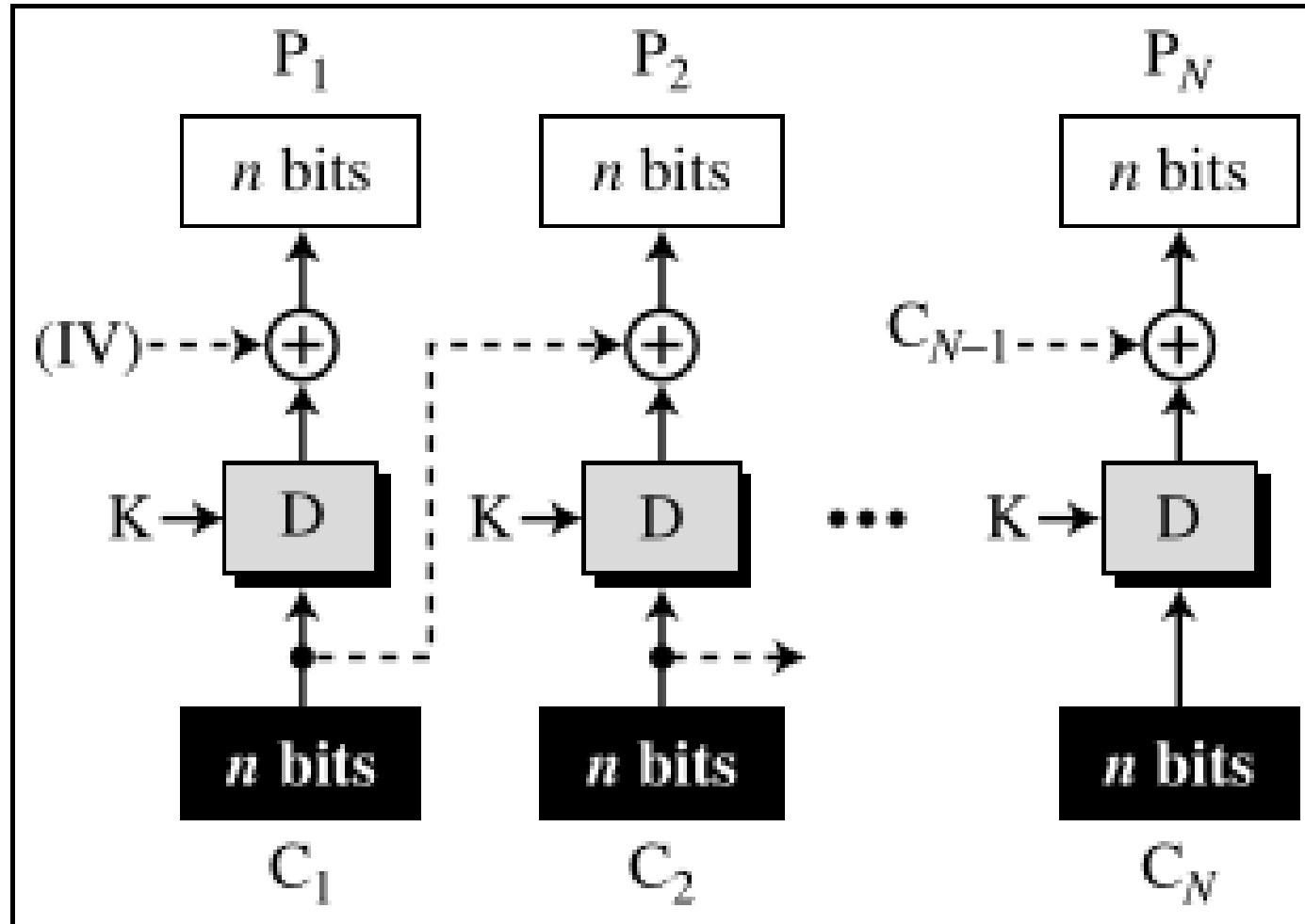
- In CBC mode, each plaintext block is XORed with the previous ciphertext block before being encrypted.
- For the first block, a phony block called the **initialization vector (IV)** is used.
- The sender and receiver agree upon a specific predetermined **IV**.
- IV can be shared as long as new IV is used for each communication.

Cipher Block Chaining



Encryption

Cipher Block Chaining



Decryption

CBC advantages and limitations

- Error in one block propagates to all other blocks.
- IV vector must be known to both sender and receiver.
- Sequential implementation.
- Cannot be parallelized.

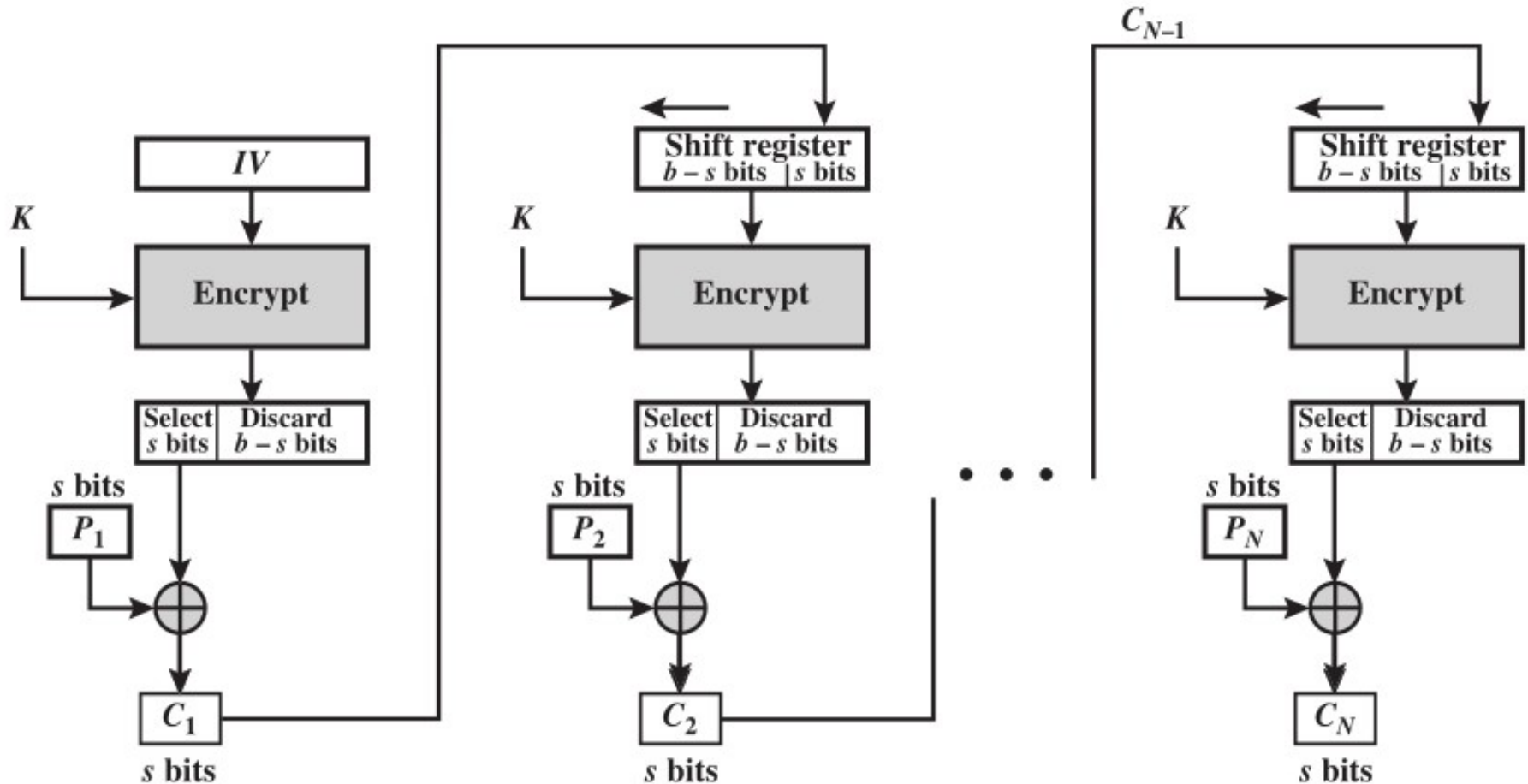
Cipher Feedback

- Cipher feedback (CFB) is one of the stream cipher.
- In stream cipher none need to pad a message.
- It also can operate in real time.
- Thus, if a character stream is being transmitted, each character can be encrypted and transmitted immediately using a character-oriented stream cipher.
- One desirable property of a stream cipher is that the ciphertext be of the same length as the plaintext.

Cipher Feedback

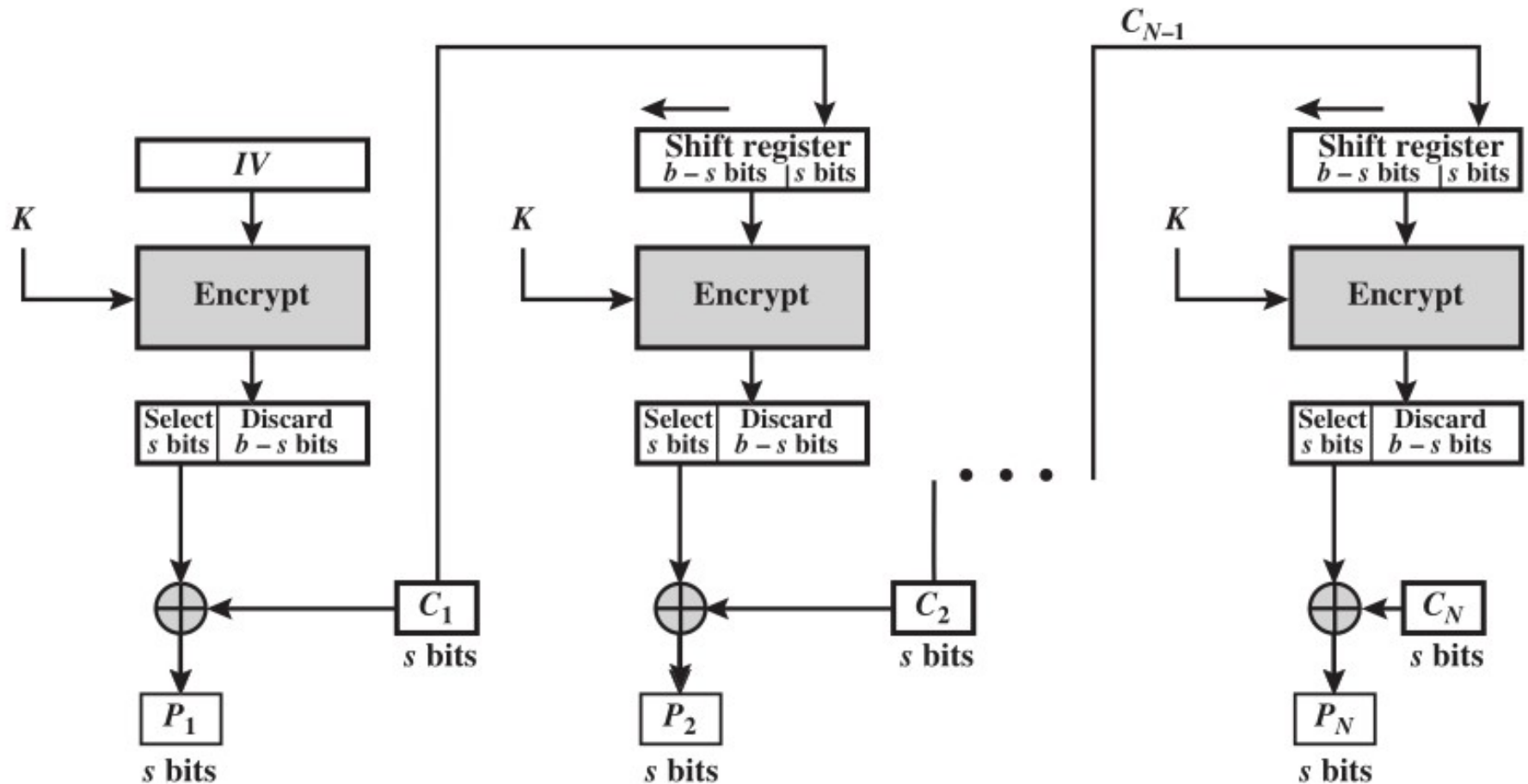
- The encryption function of the first block takes b bit IV and Key K .
- The most significant s -bits of the result is XORed with s -bit plaintext to generate s -bit ciphertext.
- The ciphertext is fed to next block.
- The input data of previous block is copied to shift register.
- The shift register is shifted by s -bits.
- Then s -bit ciphertext is appended with shift register.
- This process continues until all plaintext units have been encrypted.

Cipher Feedback



(a) Encryption

Cipher Feedback

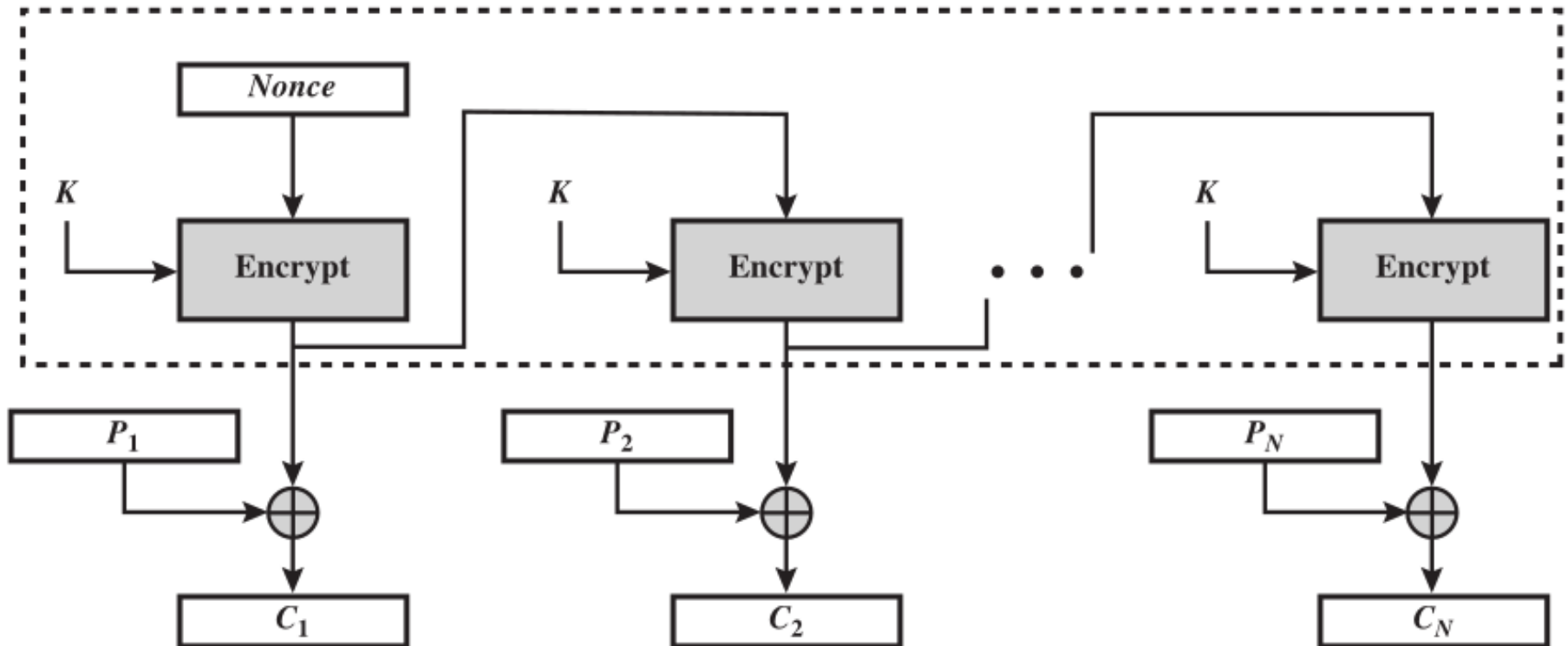


(b) Decryption

Output Feedback

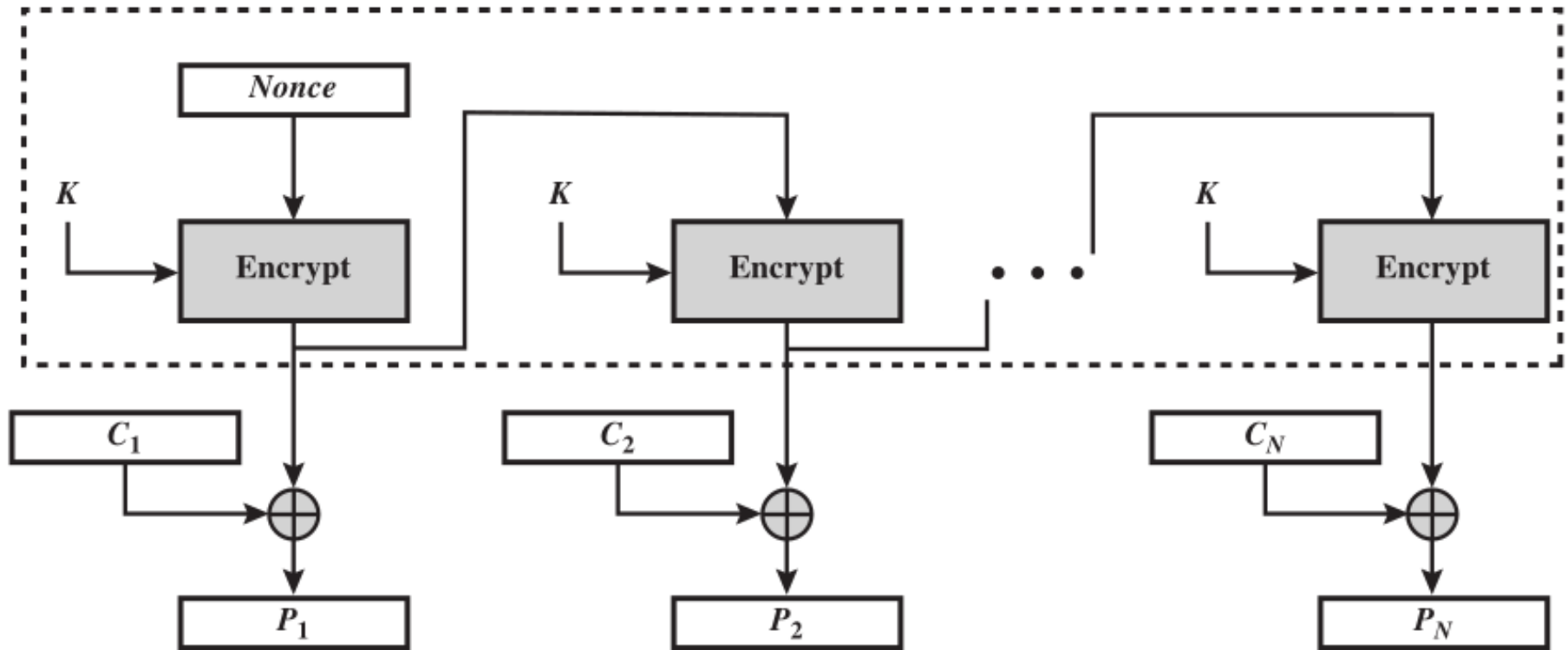
- It is similar to the Cipher feedback with simple modification.
- The output of encryption function is feed back to the next block not the cipher.
- So, next block ciphertext is independent of previous block plaintext or ciphertext.
- So, error in plaintext or ciphertext does not propagate to next block.
- One more advantage is the feedback can be precomputed.
- CS526 So, this mode works as Topic 1 stream cipher perfectly.

Output Feedback



(a) Encryption

Output Feedback



(b) Decryption

Output Feedback

- IV must be nonce means IV must be unique to each execution of the encryption operation.
- Because output of the encryption function depends only on IV and the key.
- This generates fixed output for fixed plaintext.
- So, it will be easy for attacker to determine the output of encryption function.
- One advantage is bit errors in transmission do not propagate.
- It is more vulnerable to message modification.

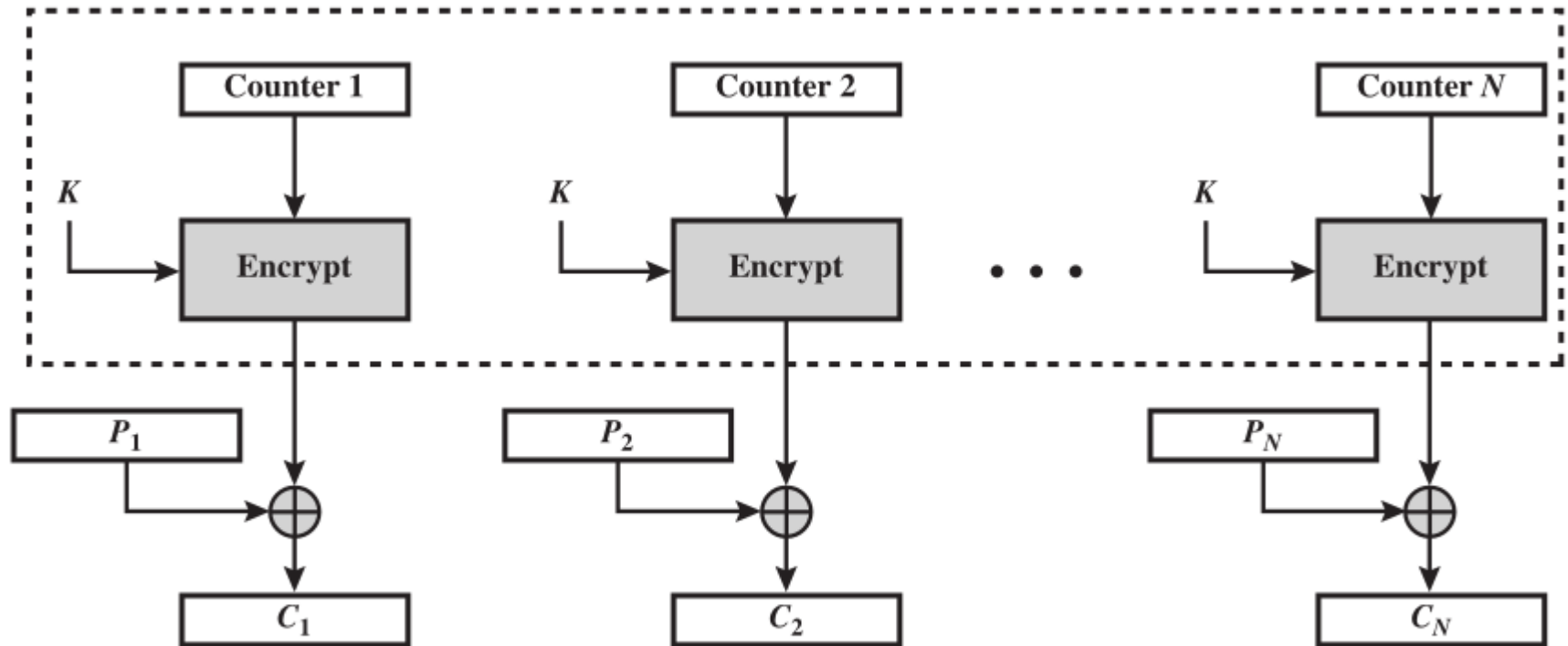
Counter Mode

- In this mode, counter equal to the plaintext block size is used.
- The counter value must be different for each plaintext block that is encrypted.
- Typically, the counter is initialized to some value and then incremented by 1 for each subsequent block (modulo 2^b , where b is the block size).
- For encryption, the counter is encrypted and then XORed with the plaintext block to produce the ciphertext block;
- there is no chaining.

Counter Mode

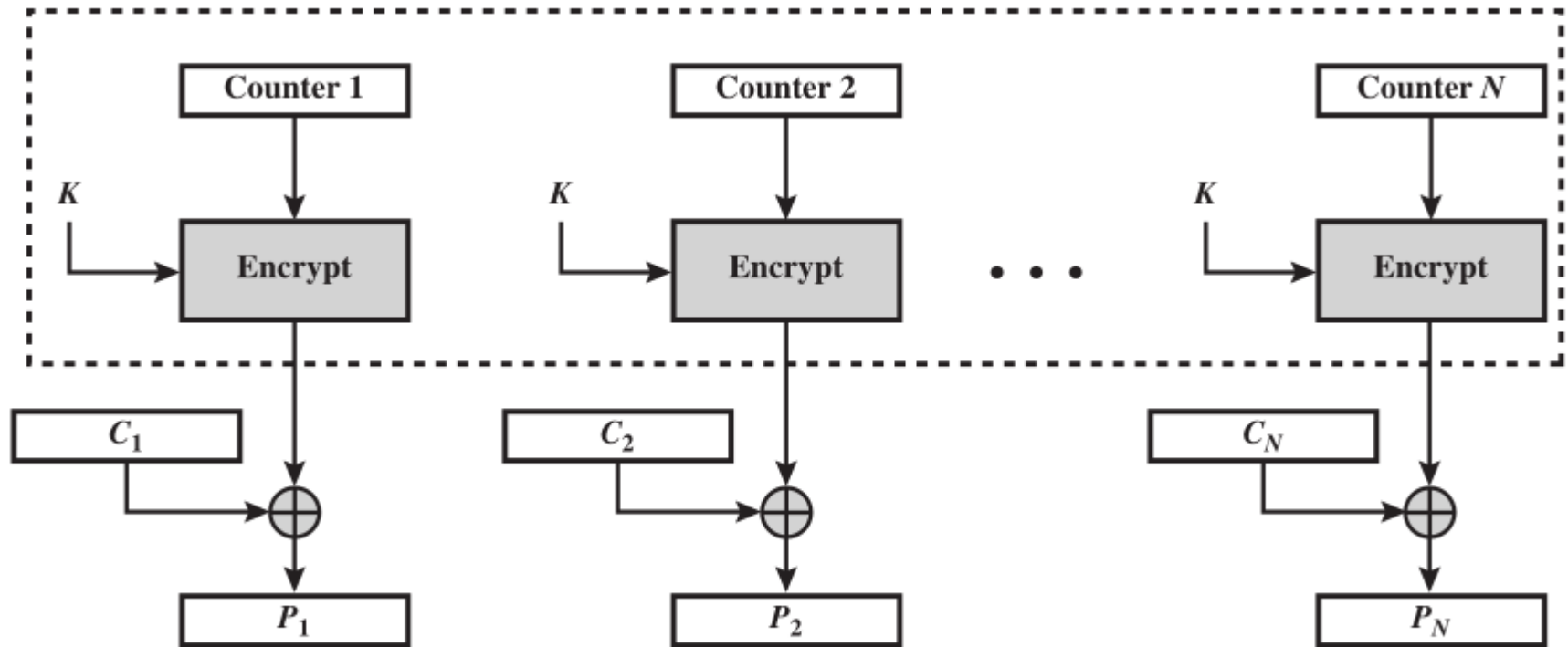
- For decryption, the same sequence of counter values is used, with each encrypted counter XORed with a ciphertext block to recover the corresponding plaintext block.
- It is more vulnerable to message modification.

Counter Mode



(a) Encryption

Counter Mode



(b) Decryption

Counter Mode

- Can do parallel encryptions in h/w or s/w
- Can preprocess in advance of need
- Random access to encrypted data blocks
- But must never reuse key/counter values, otherwise could break